

Shengfan Cao

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Summary

PhD researcher with 4+ years of hands-on experience in motion planning, trajectory optimization, and safe learning for autonomous vehicles, spanning production solver deployment to novel constrained-RL research. Seeking to bring this combination of optimization, controls, and learning expertise to industry AV systems.

Skills

- **Programming:** Python, C/C++, MATLAB, SQL.
- **Software:** PyTorch, NumPy, Pandas, ROS, CARLA, CasADi, Acados, Linux/bash, Docker, Git, Pytest, Bazel, NVIDIA Jetson.

Education

University of California, Berkeley Aug 2022 — May 2027
Ph.D. in Mechanical Engineering

- **Coursework:** Advanced Control Theory and Systems, Model Predictive Control, Nonlinear Systems, Convex Optimization, Deep Learning, Reinforcement Learning.

Tsinghua University Aug 2017 — Jun 2022
B.S. in Mechanical Engineering; B.A. in Japanese Language and Literature

- **Thesis:** Path Generation for Rapid Filling of Planar Areas with Optimized Parallel Curves
- **Award:** Scholarship for Academic Excellence (Top 10%) *sponsored by AEON*

Work Experience

Didi Research America May 2026 — Aug 2026
Motion Planning Engineer, Intern

- Re-architected the trajectory optimization core, within a large legacy C++ production planner, into an efficient augmented-Lagrangian DDP solver, cutting solve time $\sim 6\times$ against the IPOPT baseline.
- Reformulated noisy trajectory cleanup as Bayesian MAP estimation and shipped a denoiser that cut jerk / lateral jerk discomfort rates to $\sim 0-4\%$ (from 8-39% baseline) while keeping ADE/FDE competitive.
- Designed an elastic constraint-relaxation scheme with L_1 -penalized slacks, letting the solver automatically decide which constraints to relax and by how much, and shipped it onboard as a no-op-safe relaxable AL-constraint costing type.
- **Collaboration:** coordinated across ML-planner, path-core, and selection teams through regular syncs and CR reviews; delivered a tech-sharing talk on the AL-DDP solver; represented the team at an external university symposium, critically evaluating a neural primal-dual solver's infeasibility-certification and warmstart-fairness claims.
- **Tech:** C++, CasADi, Acados / HPIPM, IPOPT, Eigen, pybind11, ROS 1, Bazel, Python.

Projects

Trusted Autonomy for Rapidly Prototyped Uncrewed Ground Vehicles, Sep 2024 — Present
 Project Lead

- Engineered a rapidly deployable autonomous ground robot capable of exploring, mapping, and navigating previously unseen terrain with minimal setup and low hardware cost for expeditionary missions.
- Developed an interpretable autonomy stack that converts multi-modal sensor data into human-readable terrain and uncertainty maps to support both autonomous decision-making and operator trust.
- Implemented online mapping and real-time replanning that adapts navigation as new obstacles and terrain features are discovered during execution.
- Developed a Python data access layer over ROS2 bag files (SQLite), enabling SQL-style queries for efficient

extraction and analysis of logged autonomy data.

- Deployed and maintained the full autonomy stack on embedded onboard compute under real-world constraints, and collaborated with NIWC Pacific human-factors researchers through regular technical reviews.
- **Tech:** PyTorch, ROS 2, SQL, SLAM, YOLO, Semantic Mapping, CasADi, Jetson Orin

Sampling-Based Constrained Policy Optimization, Principal Researcher

Sep 2025 — Dec 2025

- Proposed a sampling-based weight-space projection framework for enforcing safety constraints during policy updates, formulating the update as a convex projection with theoretical guarantees that it preserves objective improvement.
- Derived a sufficient stability condition linking weight updates to closed-loop constraint satisfaction, and empirically demonstrated complete rejection of harmful supervision under adversarial experts.
- **Tech:** PyTorch, Convex Optimization, Reinforcement Learning, Safe Learning.

Safe Imitation Learning at Handling Limits, Principal Researcher

Nov 2024 — Feb 2025

- Designed a constraint-aware imitation learning framework with an actor-critic structure and reachability-based safety filters for explicit, generalizable safety modeling.
- Improved training efficiency by 150% and reduced collision rates by 60% at the vehicle's dynamic handling limits in CARLA simulation.
- **Tech:** PyTorch, CasADi, CARLA, RL, MPC, optimization, safety-aware learning.

Publications

- **Cao, S.**, Joa, E., & Borrelli, F. (2025). *A Simple Approach to Constraint-Aware Imitation Learning with Application to Autonomous Racing* [🔗](#), Oral presentation at IEEE IROS 2025.
- **Cao, S.**, Borrelli, F., & Joa, E. (2025). *Constrained Policy Optimization via Sampling-Based Weight-Space Projection* [🔗](#). Oral presentation at IFAC WC 2026.
- **Cao, S.**, & Liu, Y. (2025). *State-Conditional Adversarial Learning: An Off-Policy Visual Domain Transfer Method for End-to-End Imitation Learning* [🔗](#). Interactive presentation at IFAC WC 2026.
- Prignoli, F., **Cao, S.**, Falcone, P., Borrelli, F. (2025). *Real-Time Regulation-Aware Game-Theoretic Motion Planning for Head-to-Head Autonomous Racing*. Under review at IEEE TCST.

Academic Services

Invited Reviewer, IEEE

Apr 2025 — Present

- Reviewed 7 submissions across IFAC Automatica, CDC 2027, IFAC World Congress 2026, IEEE CASE 2025, and IEEE RA-L, upon invitation from the program committee.

Supervisor for Undergraduate Research Program, UC Berkeley

Jan 2025 — Present

- Led a team of four undergraduate students on vision-based off-road vehicle dynamics modeling and path planning, guiding research efforts, technical documentation, and successful interim presentations to faculty and industry stakeholders.

Graduate Student Instructor, UC Berkeley

Jan 2024 — May 2025

- Led discussions and office hours for 40-student sessions, and developed labs/exams on Python, MATLAB, and numerical methods for advanced engineering classes.
- Built a vehicle-dynamics simulator (dynamic bicycle model, gymnasium API) with a CARLA connector, used as course infrastructure. [🔗](#)